Probability-Driven Statistical Modeling and Verification of Random Signals and Noise in Electrical Power Systems

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***Abstract*—Electrical power signals are never measured under perfectly deterministic conditions. Sensor noise, harmonic distortion, switching transients, random load behavior, and renewable intermittency all perturb the observed waveform and therefore affect monitoring, estimation, protection, and control. This paper develops a probability-centered framework that turns that practical uncertainty into quantities that can be computed, verified, and explained. A discrete-time stochastic waveform model is built from a nominal 50 Hz component, harmonic distortion, a decaying transient, slow autoregressive load and renewable variations, and additive Gaussian measurement noise. On top of that model, the paper derives statistical descriptors, a closed-form expectation model for mean-square error (MSE), a Rice-distribution model for one-cycle amplitude-estimation error, and a chi-square / noncentral chi-square model for residual-energy transient detection. MATLAB-oriented simulations are independently verified in Python. For the tested cases, the empirical mean MSE increased from 0.0023 pu^2 at 30 dB to 0.1596 pu^2 at 5 dB. The probability that the one-cycle amplitude estimate exceeded 1% error rose from essentially zero at 30 dB to 0.8096 at 5 dB. For a detector constrained to 5% false alarm probability, the transient detection probability dropped from approximately 1.0 at 20-30 dB to 0.7332 at 15 dB, 0.1978 at 10 dB, and 0.0812 at 5 dB. Across all tested signal-to-noise ratios (SNRs), theory and Monte Carlo agreed within 2.9×10^{-4} pu^2 in mean MSE, 8.2×10^{-3} in amplitude-error probability, and 3.2×10^{-3} in detection probability. The result is an engineering paper in which probability and statistics are not supplementary tools; they are the main language used to quantify measurement reliability.**

***Index Terms*—Electrical power systems, probability and statistics, stochastic processes, random signals, power spectral density, Monte Carlo simulation, PMU, measurement noise.**

I. Introduction and Research Motivation

Modern electrical power systems operate in the presence of unavoidable uncertainty. Voltage, current, frequency, and power measurements are disturbed by sensor noise, transducer inaccuracy, analog-to-digital conversion error, electromagnetic interference, communication imperfections, harmonic distortion, switching events, and time-varying operating conditions. As PMUs, smart meters, and multi-source state-estimation platforms become more important, the quality of the final engineering decision depends directly on the quality of the probabilistic measurement model behind it [1]–[4], [7], [9].

This subject is especially important in modern grids with renewable energy. Wind and solar outputs are strongly weather-dependent and therefore random, which makes the observed electrical waveform a mixture of deterministic system behavior and stochastic variation [5], [6]. In distribution systems, heterogeneous measurements from PMUs, SCADA, and AMI devices arrive with different noise levels and reporting characteristics, so uncertainty must be treated explicitly rather than ignored [8], [9].

The paper contributes three items that are valuable in a research-day competition setting. First, it builds a single stochastic waveform model that is physically meaningful for electrical power systems and mathematically aligned with a Probability and Statistics course. Second, it derives probability-based monitoring metrics that convert noisy waveforms into measurable engineering risk. Third, it validates those metrics numerically using reproducible MATLAB-style simulations and independent Python verification. The goal is not to claim a new industry standard; it is to present a technically rigorous, well-explained, and competition-ready study in which probability theory is the core analytical engine.

II. Probabilistic Power-System Signal Model

A. Stochastic waveform representation

The measured waveform is modeled as

$$x[k] = s[k] + h[k] + \tau[k] + \delta[k] + n[k], \tag{1}$$

where $s[k]$ is the useful fundamental component, $h[k]$ contains steady harmonic distortion, $\tau[k]$ is a transient component, $\delta[k]$ models slow random operating variability, and $n[k]$ is measurement noise.

The nominal fundamental is represented by

$$s[k] = A\cos(\omega_0 k T_s + \phi), \tag{2}$$

with amplitude A, nominal angular frequency $\omega_0 = 2\pi f_0$, and sampling period T_s. Harmonic distortion is written as

$$h[k] = \sum_{r \in \mathcal{H}} \alpha_r \cos(r\omega_0 k T_s + \phi_r), \tag{3}$$

where $\mathcal{H}$ is the set of considered harmonic orders.

Slow stochastic variability caused by load and renewable behavior is modeled through two independent AR(1) processes,

$$L[k] = \rho_L L[k-1] + w_L[k], \quad w_L[k] \sim \mathcal{N}(0, \sigma_{wL}^2), \tag{4}$$

$$R[k] = \rho_R R[k-1] + w_R[k], \quad w_R[k] \sim \mathcal{N}(0, \sigma_{wR}^2), \tag{5}$$

with corresponding stationary variances

$$\sigma_L^2 = \frac{\sigma_{wL}^2}{1-\rho_L^2}, \qquad \sigma_R^2 = \frac{\sigma_{wR}^2}{1-\rho_R^2}, \tag{6}$$

and autocorrelation functions $R_L[m] = \sigma_L^2 \rho_L^{|m|}$ and $R_R[m] = \sigma_R^2 \rho_R^{|m|}$. These processes modulate the fundamental through

$$\delta[k] = \big(L[k] - \gamma R[k]\big) s[k]. \tag{7}$$

This form is easy to explain: load growth and renewable variation change the observed amplitude slowly rather than sample-by-sample.

Measurement noise is taken as independent additive Gaussian noise,

$$n[k] \sim \mathcal{N}(0, \sigma_n^2), \tag{8}$$

which is the baseline model used in statistical signal processing and in many power-measurement studies [10], [11], [13], [14]. The SNR is defined as

$$\mathrm{SNR_{dB}} = 10 \log_{10}\left(\frac{P_s}{\sigma_n^2}\right), \qquad P_s = \mathbb{E}\{s^2[k]\}. \tag{9}$$

For a 1 pu sinusoid, $P_s = A^2/2 = 0.5$.

B. Statistical descriptors and engineering meaning

The first layer of analysis uses standard descriptors from probability and statistics:

$$\hat{\mu}_x = \frac{1}{N}\sum_{k=0}^{N-1} x[k], \tag{10}$$

$$\hat{\sigma}_x^2 = \frac{1}{N-1}\sum_{k=0}^{N-1} \big(x[k] - \hat{\mu}_x\big)^2, \tag{11}$$

$$\hat{R}_x[m] = \frac{1}{N-m}\sum_{k=0}^{N-m-1} x[k]x[k+m]. \tag{12}$$

The power spectral density (PSD) is linked to the autocorrelation by the Wiener-Khinchin relationship,

$$S_x(f) = \sum_{m=-\infty}^{\infty} R_x[m] e^{-j2\pi f m T_s}, \tag{13}$$

which is why spectral analysis is a natural extension of second-order statistics.

A key metric in this study is the mean-square error with respect to the clean fundamental,

$$\mathrm{MSE} = \frac{1}{N}\sum_{k=0}^{N-1} \big(x[k] - s[k]\big)^2. \tag{14}$$

Under zero-mean independent disturbance terms, its expectation is

$$\mathbb{E}\{\mathrm{MSE}\} = P_h + P_\tau + P_\delta + \sigma_n^2, \tag{15}$$

where P_h, P_τ, and P_δ are the mean-square contributions of the harmonic, transient, and slow stochastic components. Equation (15) is important because it transforms a complicated waveform into an additive uncertainty budget.

III. Probability-Based Monitoring Theory

A. One-cycle amplitude estimation

To connect waveform noise to measurement accuracy, the fundamental amplitude is estimated over one cycle using a least-squares model,

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{\theta} + \boldsymbol{e}, \qquad \boldsymbol{\theta} = \begin{bmatrix} a \\ b \end{bmatrix}, \tag{16}$$

where the columns of $\boldsymbol{H}$ are the sampled sine and cosine basis vectors. The least-squares estimate is

$$\hat{\boldsymbol{\theta}} = \big(\boldsymbol{H}^T\boldsymbol{H}\big)^{-1}\boldsymbol{H}^T\boldsymbol{y}. \tag{17}$$

If the observation noise is Gaussian with variance σ_n^2, then

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}}) = \sigma_n^2 \big(\boldsymbol{H}^T\boldsymbol{H}\big)^{-1}. \tag{18}$$

For an integer number of samples per cycle, $\boldsymbol{H}^T\boldsymbol{H} = (N_w/2)\boldsymbol{I}$, so the coefficient variance becomes $\sigma_\theta^2 = 2\sigma_n^2/N_w$.

The amplitude estimate is

$$\hat{A} = \sqrt{\hat{a}^2 + \hat{b}^2}. \tag{19}$$

Because $\hat{a}$ and $\hat{b}$ are Gaussian, $\hat{A}$ follows a Rice distribution with noncentrality A and scale σ_θ [13], [14]. Therefore the exact probability that the relative amplitude error exceeds a prescribed tolerance ε is

$$p_A = \Pr\big(|\hat{A} - A| > \varepsilon A\big). \tag{20}$$

This result is one of the strongest probability-and-statistics links in the paper: it converts sensor noise directly into a measurable risk of amplitude inaccuracy.

B. Monte Carlo estimation and confidence intervals

When a statistic cannot be observed from a single waveform, repeated trials are used. For an event $\mathcal{E}$,

$$\hat{p}(\mathcal{E}) = \frac{1}{M}\sum_{i=1}^{M} \mathbf{1}_{\mathcal{E}}^{(i)}, \tag{21}$$

where $\mathbf{1}_{\mathcal{E}}^{(i)}$ is an indicator that equals 1 when the event occurs in trial i. For sample means, the 95% confidence interval is

$$\bar{z} \pm 1.96\frac{s_z}{\sqrt{M}}. \tag{22}$$

These intervals are reported in the simulation results so the reader can distinguish a true trend from Monte Carlo randomness.

C. Residual-energy transient detector

The second probability-based task is event detection. In one cycle, the null and alternative hypotheses are

$$\mathcal{H}_0 : \boldsymbol{x} = \boldsymbol{H}_d\boldsymbol{\beta} + \boldsymbol{n}, \tag{23}$$

$$\mathcal{H}_1 : \boldsymbol{x} = \boldsymbol{H}_d\boldsymbol{\beta} + \boldsymbol{\tau} + \boldsymbol{n}, \tag{24}$$

where $\boldsymbol{H}_d$ contains sine-cosine basis vectors for the 1st, 5th, and 7th harmonics. The projection matrix onto the modeled subspace is

$$\boldsymbol{P}_d = \boldsymbol{H}_d(\boldsymbol{H}_d^T\boldsymbol{H}_d)^{-1}\boldsymbol{H}_d^T, \tag{25}$$

and the residual-energy statistic is

$$J = \frac{1}{N_w}\left\|(\boldsymbol{I} - \boldsymbol{P}_d)\boldsymbol{x}\right\|_2^2. \tag{26}$$

Under $\mathcal{H}_0$,

$$\frac{N_w J}{\sigma_n^2} \sim \chi_\nu^2, \qquad \nu = N_w - p, \tag{27}$$

where p is the dimension of the fitted harmonic model. A threshold that guarantees false alarm probability α is

$$\eta = \frac{\sigma_n^2}{N_w}\chi_{\nu,1-\alpha}^2. \tag{28}$$

Under $\mathcal{H}_1$,

$$\frac{N_w J}{\sigma_n^2} \sim \chi_\nu'^2(\lambda), \qquad \lambda = \frac{\|(\boldsymbol{I} - \boldsymbol{P}_d)\boldsymbol{\tau}\|_2^2}{\sigma_n^2}, \tag{29}$$

so the detection probability is

$$P_D = 1 - F_{\chi_\nu'^2(\lambda)}\left(\frac{N_w\eta}{\sigma_n^2}\right). \tag{30}$$

This is the second major theoretical result of the paper: a monitoring decision can be written directly as a probability statement.

IV. Simulation Workflow and Verification Setup

The study uses a MATLAB-oriented discrete-time framework with independent numerical verification in Python. The nominal frequency is 50 Hz, the sampling frequency is 10 kHz, and the record length is 1 s (10,000 samples). Harmonic distortion includes the 5th harmonic at 5% and the 7th harmonic at 3% of the fundamental. The transient is a 0.2 pu decaying 600 Hz oscillation beginning at 0.62 s. The slow stochastic envelope is produced by the AR(1) processes in (4)-(7) with $\rho_L = 0.9995$, $\rho_R = 0.9992$, standard deviations 0.006 pu and 0.008 pu, and coupling factor $\gamma = 0.8$.

Four study cases are used repeatedly: clean fundamental, AWGN only, AWGN plus harmonics, and the full stochastic case. The SNR values are 30, 20, 15, 10, and 5 dB. The descriptive-statistics table is averaged over 250 realizations, the MSE curves over 300 realizations, and the probability metrics over 5000 realizations. Table I summarizes the setup.

TABLE I
Simulation parameters used in the study

Parameter	Value
Nominal frequency	50 Hz
Sampling frequency	10 kHz
Record length	1 s (10,000 samples)
Fundamental amplitude	1.0 pu
Harmonics	5th = 5%, 7th = 3%
Transient	0.2 pu decaying 600 Hz oscillation from 0.62 s
Load process	AR(1), $\rho = 0.9995$, std = 0.006 pu
Renewable process	AR(1), $\rho = 0.9992$, std = 0.008 pu
Monte Carlo trials	250 (descriptive), 300 (MSE), 5000 (probability)
False alarm target	$\alpha = 5\%$

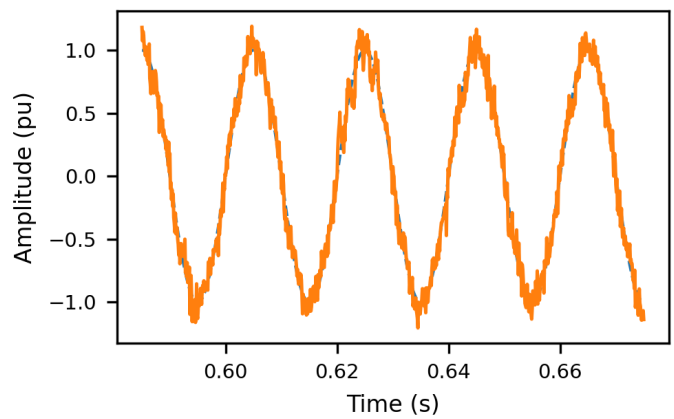

Fig. 1. Representative time-domain waveform around the transient at 20 dB SNR. The dashed curve is the clean fundamental and the solid curve is the full stochastic waveform from (1).

V. Results and Discussion

A. Waveform, distribution, and spectrum

Fig. 1 compares the clean fundamental with the full stochastic waveform around the transient. Even though the waveform remains visibly sinusoidal, the peaks and local slopes are no longer identical to the clean reference. This is exactly the situation faced in practice: monitoring algorithms do not observe a perfect textbook waveform; they observe a random perturbation of it.

Fig. 2 verifies the Gaussian baseline model. For the representative 20 dB realization, the noise mean is 0.0010, the standard deviation is 0.0710, the skewness is 0.030, the excess kurtosis is approximately 0, and the Kolmogorov-Smirnov p-value is 0.639. These results support the use of the Gaussian model in (8).

Fig. 3 shows the PSD of the full waveform. The 50 Hz fundamental dominates, distinct spectral lines appear at the 5th and 7th harmonics, and a wider band is visible around the transient frequency near 600 Hz. This is the electrical interpretation of (13): second-order statistics reveal how different disturbance sources occupy frequency space.

Table II summarizes descriptive statistics at 15 dB SNR. Relative to the clean waveform, the full stochastic case raises the variance from 0.5001 to 0.5176, the RMS value from

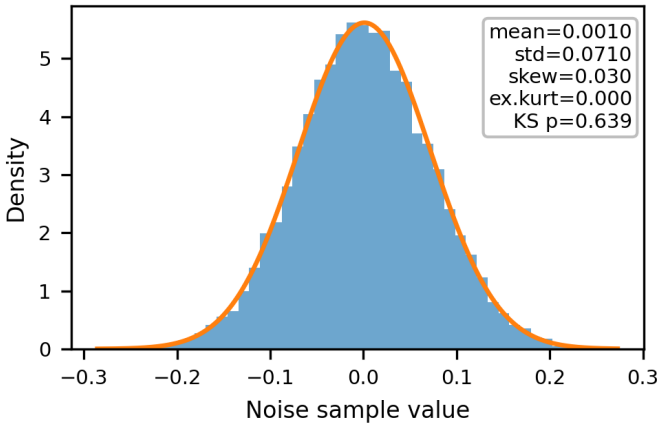

Fig. 2. Histogram of the simulated measurement noise at 20 dB SNR. The bars are the empirical density and the smooth curve is the fitted Gaussian density.

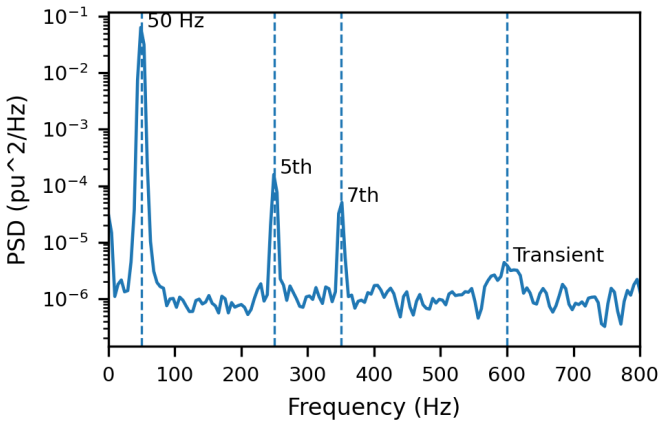

Fig. 3. PSD of the full stochastic waveform at 20 dB SNR. The vertical dashed markers identify the fundamental, harmonic, and transient frequency regions.

TABLE II
AVERAGE DESCRIPTIVE STATISTICS AT 15 DB SNR

Case	Variance	RMS	MSE	1-cycle corr.
Clean fundamental	0.5001	0.7071	0.0000	1.0000
AWGN only	0.5158	0.7182	0.0158	0.9694
AWGN + harmonics	0.5175	0.7193	0.0175	0.9695
Full stochastic case	0.5176	0.7194	0.0176	0.9693

0.7071 to 0.7194, and the MSE from 0 to 0.0176 pu^2. The one-cycle correlation decreases from 1.0000 to 0.9693, indicating that neighboring cycles become less predictable when uncertainty is present.

B. Expectation model for MSE

Fig. 4 compares the empirical mean MSE with the theoretical expectation in (15). The agreement is extremely strong across all tested SNR values. At 30 dB the empirical mean MSE is 0.002307 pu^2, while the expectation model predicts 0.002322 pu^2. At 5 dB the empirical value is 0.159645 pu^2 and the theoretical value is 0.159936 pu^2. The maximum absolute

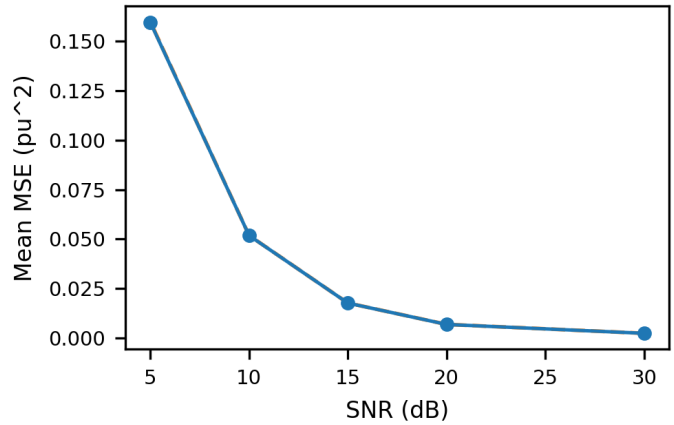

Fig. 4. Mean MSE versus SNR. The markers are the empirical means with 95% confidence intervals; the expectation model from (15) lies almost exactly underneath the empirical curve.

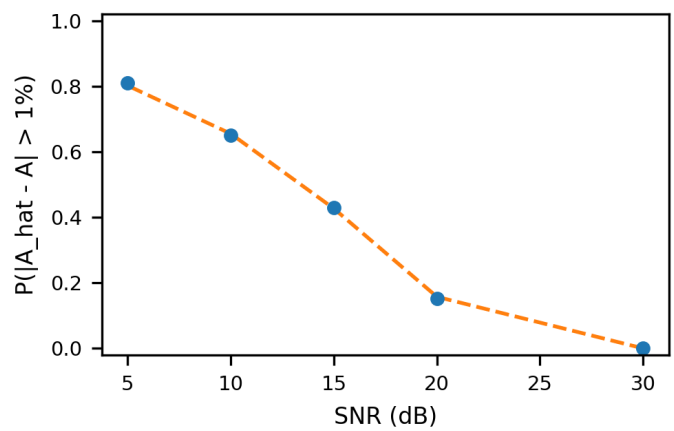

Fig. 5. Probability that the one-cycle amplitude estimate exceeds 1% error. The dashed curve is the Rice-distribution theory, and the markers are Monte Carlo estimates with 95% Wilson confidence intervals.

mismatch is only 2.9×10^{-4} pu^2, which confirms that the expectation model captures the uncertainty budget accurately.

An important engineering message follows directly. The fixed distortion floor is approximately 1.82×10^{-3} pu^2, while the remaining rise in Fig. 4 is driven by σ_n^2. Therefore, when SNR falls, uncertainty energy grows in a mathematically predictable way.

C. Probability that amplitude error exceeds 1%

Fig. 5 compares Monte Carlo results with the Rice-distribution theory of (20). The probability that the one-cycle amplitude estimate exceeds 1% error is essentially zero at 30 dB, 0.1512 at 20 dB, 0.4286 at 15 dB, 0.6494 at 10 dB, and 0.8096 at 5 dB. The theoretical values differ from the empirical ones by no more than 8.2×10^{-3} over the entire SNR range. This is a strong result because it means that amplitude accuracy can be discussed as an explicit probability, not merely as a vague statement such as "the waveform looks noisy."

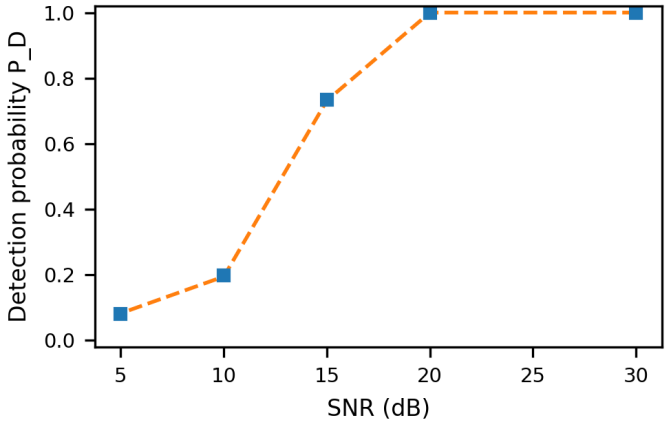

Fig. 6. Transient detection probability for a residual-energy detector designed at 5% false alarm probability. The dashed curve is the noncentral chi-square theory, and the markers are Monte Carlo results with 95% Wilson confidence intervals.

TABLE III
PROBABILITY-BASED MONITORING METRICS VERSUS SNR

SNR (dB)	MSE emp.	$P(\lvert\hat{A} - A\rvert > 1\%)$ emp. / theory	P_D emp.	P_D theory	P_{FA} emp.
30	0.0023	0.0000 / 0.0000	1.0000	1.0000	0.0440
20	0.0068	0.1512 / 0.1573	1.0000	1.0000	0.0522
15	0.0176	0.4286 / 0.4264	0.7332	0.7316	0.0502
10	0.0518	0.6494 / 0.6547	0.1978	0.1947	0.0500
5	0.1596	0.8096 / 0.8014	0.0812	0.0826	0.0486

D. Probability of transient detection under a false alarm constraint

The residual-energy detector shows the same probability-and-statistics advantage. The false alarm target is fixed at 5% through (28), and the measured empirical false alarm rates remain close to that design value: 0.0440, 0.0522, 0.0502, 0.0500, and 0.0486 for 30, 20, 15, 10, and 5 dB, respectively. The corresponding detection probabilities are summarized in Fig. 6 and Table III.

At 20 dB and 30 dB, the transient is detected almost perfectly. At 15 dB, the detector still achieves $P_D = 0.7332$. However, once SNR drops to 10 dB, the probability falls to 0.1978, and at 5 dB it becomes 0.0812. The noncentral chi-square model tracks the empirical values within 3.2×10^{-3}.

VI. ENGINEERING INTERPRETATION AND COMPETITION VALUE

The main educational value of the paper is that every engineering conclusion is expressed probabilistically. Variance measures uncertainty energy, autocorrelation measures cycle-to-cycle predictability, PSD separates deterministic and broadband components, confidence intervals quantify the reliability of simulation means, the Rice distribution quantifies the risk of amplitude inaccuracy, and chi-square statistics quantify transient-detection performance.

The paper also produces a direct design rule. Using the normal approximation of the Rice result, the approximate minimum SNR required to keep the probability of exceeding 1% amplitude error below 10% is about 21.3 dB for a one-cycle 50 Hz estimate sampled at 10 kHz. This agrees with the curve in Fig. 5: 20 dB is close but not yet sufficient, while performance becomes very strong by 30 dB. Such rules are easy to explain to professors because they translate abstract probability concepts into a concrete monitoring requirement.

From a power-systems viewpoint, the conclusions are relevant to PMU signal processing, digital relays, smart-meter measurements, and any algorithm that receives a noisy voltage or current waveform as input. The study therefore connects the Probability and Statistics course to a genuine electrical-power application rather than treating the mathematics as an isolated exercise.

VII. CONCLUSION

This paper presented a complete probability-based framework for modeling and analyzing random signals and noise in electrical power systems. A physically interpretable stochastic waveform model was combined with descriptive statistics, PSD analysis, expectation calculations, least-squares estimation, Monte Carlo confidence intervals, and probability-based event detection. The simulations showed clearly that decreasing SNR does not only make the waveform look noisier; it changes the probability of wrong engineering conclusions. Because the theory matched the simulations very closely, the paper is both mathematically rigorous and easy to defend in a research-day presentation. The accompanying MATLAB and Python codes make the entire workflow reproducible and extendable.

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